

Yu Hsuan Chen (Tina)

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Solution Architect / SRE / DevOps

✉ tinachen1996121@gmail.com · 📞 0917-577-377 · 📍 New Taipei City, Taiwan
🔗 cake.me/tinachen1996121

PROFILE

7-year cross-cloud SRE / DevOps engineer with hands-on production experience across **AWS, GCP, and on-premise** environments. Background spans crypto exchange (high regulatory bar), martech SaaS, hardware / IoT, and education tech — covering **SOC 2 / ISO 27001 compliance, org-scale cloud migration, security incident response, cost optimization, and AI-powered SRE tooling**. Currently driving SOC 2 readiness, pre-production environment design, and an LLM-based SRE agent at ViewSonic.

HIGHLIGHTS

- **Multi-cloud** — AWS (~5 yrs) + GCP (2 yrs); 11+ AWS accounts, multi-region production
- **Compliance** — SOC 2 audit readiness · ISO 27001 monitoring
- **Migration scale** — GCP Org migration (Shared VPC, PSC, DMS) · GitHub org migration (69 repos)
- **Security** — Drove breach response & full EKS rebuild · Secret Manager + ESO + key rotation · WAF / GuardDuty / SCC
- **Cost** — AWS Savings Plans + RI (RDS / Redis / OpenSearch + Compute SP) · GCP CUD · 100% Spot GKE · KubeCost
- **AI / automation** — Built OpenClaw (LLM-based SRE agent) · 30+ Claude Code skills + hook-driven automation adopted by infra team

EXPERIENCE

DevOps Engineer · ViewSonic Corporation

New Taipei · 2025 – Present · DEV Department

🛡️ SOC 2 Audit Readiness

- Drove SOC 2 documentation preparation (PBC mapping) across RD / SRE / QA, coordinating evidence collection from 3 functions.
- Implemented strict IAM policies, MFA enforcement, and automated account review workflows.
- Enforced Dev / Test / Prod environment isolation and secure SDLC practices with mandatory code reviews.
- Managed cloud asset inventory and automated data encryption via AWS KMS / Azure Key Vault.
- Authored Business Continuity (BCP) and Disaster Recovery (DR) plans meeting RTO / RPO targets.

- Configured centralized audit logging, vulnerability scanning, and automated alerting systems.

Pre-Production (RC) Environment

- Stood up release-candidate environment **inside the production AWS account**, sharing EKS cluster / VPC / ACM while isolating data and compute — cuts infra spend versus full prd-duplicate setup.
- Separated RDS / Redis / S3 with `-rc` suffix, including data migration, firewall rules, and dedicated DB credentials.
- Created dedicated CloudFront distributions for `*rc.classswift.com` with Lambda@Edge, reusing the same WAF and Istio NLB.
- Split EKS node pools and namespaces via ArgoCD `value.rc.yaml` with RC node selectors.
- Isolated Parameter Store keys under `/rc/` path for both infra and app config.
- Extended Grafana dashboards and Kibana indices to cover RC alongside prd.

OpenClaw — AI SRE Agent

- Built LLM-based SRE assistant that runs AWS CLI / kubectl across dev / stg / prd accounts; designed **3-layer Terraform state architecture** with S3 backend and cross-account assume-role.
- Deployed on a single EC2 running k3s, connected via SSM through VPC Endpoints (no public IP, no inbound exposure).
- Slack integration via API Gateway + Lambda with DynamoDB-backed conversation memory.
- Designed modular **skill system (8 skills)** loaded on demand via K8s ConfigMap.
- Packed internal repos into searchable XML references so the LLM can look up infra configs during incidents.
- Wrote custom system prompts with incident triage SOP and guardrails to keep the bot action-oriented.

GitHub Organization Migration & Governance

- Drove migration of **69 repositories** from VSX-ViewSonic to Viewsonic-EDU (Apr 2026), standardizing branch protection, ruleset, and `delete_branch_on_merge` baseline via gh API automation. Zero downtime, zero broken CI pipelines.
- Established org-wide governance baseline: ruleset templates, bypass approval rules, PR auto-delete-branch — applied uniformly across all migrated repos.

Multi-Agent Automation Platform (*internal DevTool*)

- Built Claude Code-based automation framework: **30+ custom skills** covering AWS infrastructure, Kubernetes, GitHub, CI/CD, and security workflows; hook system for event-driven actions (pre-commit / post-edit / on-PR); 3 parallel execution modes (team-mode / parallel-research / parallel-dev).

- Packed internal AWS / Terraform / K8s reference docs into searchable XML knowledge bases for LLM lookup during incident response and IaC review.
- Hook-driven event automation: PR open / merge triggers, post-edit linting, pre-commit validation pipelines — eliminated manual hand-offs in daily ops.
- Adopted by infra team as daily-driver tooling.

Cross-Account Networking & Alerting

- Designed VPC peering (vsx-dev ↔ sss-ai-dev) including TGW attachment cleanup, RAM share, and route table propagation — unblocked AI team's cross-account service access in 1 sprint.
- Built reusable Terraform module (`aws-alert` repo) for RDS / ElastiCache / EKS health → SNS → on-call rotation; consolidated 20+ ad-hoc CloudWatch alarms into one declarative source of truth.
- Operate AWS Client VPN (dev / prd endpoints) with SNAT routing for engineer access to internal resources (EKS, RDS, ElastiCache).

AWS Cost & IAM Identity Center Governance

- Integrated AWS Cost Explorer at **Org payer level** (admin account, us-east-1) for cross-account **NetAmortizedCost** reporting with RI / Savings Plan amortization across 11 accounts; replaced per-account UnblendedCost guesswork that understated true monthly cost by **20–35%**.
- Centralized IAM Identity Center governance (us-east-1): standardized shared permission sets (`vsx-fullstack-v1` / `vsx-ro-v1` / `vsx-admin-v1` / `vsx-vendor-v1`) across all subaccount families instead of per-app proliferation — prevents policy drift and simplifies audit.
- Automated user / group / permission-set provisioning via aws-cli + Identity Store + SSO-Admin APIs (modular bash workflow).

Security Scan Automation (SCA / SAST)

- Integrated central reusable GitHub Actions workflow (`edu-security-lab-va`) with **DefectDojo** + **Dependency-Track** for SBOM generation and vulnerability scanning across all migrated repos.
- OIDC-based AWS role assumption (`DevSecOps-Github-OIDC-bot`) for secure cross-account scan access; standardized `BU/product/repo-name` taxonomy for portfolio classification.
- Authored integration playbook covering common gotchas (workflow permissions, OIDC token scope, DT project ACL setup) — reduced first-time integration failures across BU teams.

SRE • BitoPro Technology

Taipei • 2023 – 2025 • O&M Department • (Taiwan's largest crypto exchange)

Security Breach Response

- Rebuilt EKS cluster from scratch and migrated all applications post-incident.
- Rotated all secrets: Secret Manager entries, EC2 user passwords, K8s Secrets.

- Re-deployed K8s resources through Jenkins pipeline; reviewed every Network Policy / Security Group inbound & outbound rule.
- Isolated payment systems onto stand-alone EC2 instances.
- Hardened with AWS Config / Security Hub / GuardDuty; restricted SRE IAM permissions to least-privilege; reviewed all WAF rules.

AWS Infrastructure as Code

- Owned production AWS CDK codebase (TypeScript).

Cost Optimization

- AWS Savings Plans + RIs across RDS / Redis / OpenSearch + Compute SP; daily budget alerts; KubeCost rollout with per-deployment CPU/mem requests / limits tuning.

Observability & Monitoring

- AWS service monitoring · PMM DB query insights · Grafana OnCall notifications · OpenTelemetry tracing · Thanos central monitoring · ELK / OpenSearch logging architecture.

Security Hardening

- Architected Secret Manager + External Secrets Operator pipeline with automatic key rotation; routed VPC Flow Logs + WAF logs to S3 for Athena reporting.
- Deployed cert-manager for TLS lifecycle; tightened Network Policies, Security Policies, NACLs; ran Chaos Monkey drills; tuned WAF rules.

CI/CD Pipelines

- Built Jenkins pipeline with dynamic pod templates (JNLP + Kaniko containers, S3 cache, EFS-mounted Go build cache, ECR-refresher helper for credential rotation).
- Mirrored architecture on GitLab Runner.

Reliability & Stability

- Pod HPA / VPA · Node ASG · JMeter stress testing · daily Velero backups (RDS / Redis / EBS / PVC) · operated multi-broker, partitioned, replicated Kafka clusters.

Operational Maintenance

- Helm chart upgrades · SSM Agent updates · EKS version upgrade workflow · routine release deploys.

SRE · awoo Intelligence

Taipei · 2021 – 2023 · Cloud Infra Department

GCP Organization Migration

- Designed and executed GCP organization migration end-to-end: Shared VPC, VPC peering, GKE / Cloud SQL / Redis IP range planning, Private Service Connect (PSC), Database Migration Service (DMS).
- Built migration playbook: Terraform IaC, ArgoCD CI/CD, MongoDB dump / restore, continuous Cloud SQL migration, GCE migration, public IP migration with **minimized downtime traffic switch**.

On-Premise Data Center Operations

- NAS backup solution · Fortinet site-to-site VPN · VLAN environment isolation (Dev / Staging / Production).

Disaster Recovery

- Planned Tokyo subnet + GKE Pod / Service IP ranges · backup / restore procedures · DR rehearsals.

GitOps CI/CD on GKE

- Terraform Infra Pipeline — GitLab + GKE runner + Helm + Terraform.
- ArgoCD Application Pipeline — GitLab + GKE ArgoCD + Kustomize + Helm.

GCP Cost Optimization

- KubeCost-driven deployment cost tuning; **100% Spot GKE**; Redis & MongoDB self-hosted on GKE; Nginx Ingress Controller replacing internal LB.
- Committed Use Discounts (CUD) for VM / Cloud SQL / Redis; BQ + GCS data lifecycle policies; daily BQ query quota limits; Cloud Logging exclusion rules; scheduled GCE stop / restart; auto-cleanup of unattached disks / external IPs.

Monitoring

- Thanos central monitoring · Cloud Monitoring central alerting · service uptime checks.

Security

- Security Command Center · VM patch / NTP / OS update automation · VM vulnerability scanner · Cloud SQL SSL enforcement · company-wide MFA · Cloud Armor WAF for L7 DDoS mitigation · firewall hardening (removed `0.0.0.0/0` source on SSH / RDP ports).

Cloud Engineer · Askey Computer Corp

New Taipei · 2019 – 2021 · Cloud Dev Department · (ASUS subsidiary, hardware / IoT)

Built cloud APIs and IoT services for smart home, smart bus, FOTA, and firmware management products.

- **Smart Shuttle Bus IoT** — Node.js, Raspberry Pi GPIO, AWS IoT Core, MQTT, WebSocket; connected smart stations, shuttles, and cloud.

- **Smart Home Capability API** — Node.js, Cloud Functions, IoT Core, PubSub, DataStore, MQTT; appliance capability sync between devices and cloud.
- **WiFi 6 Router Management Report API** — Node.js, BigQuery, Cloud Functions, API Gateway; account registration + device onboarding analytics.
- **AI Search Service API** — Python, S3, Lambda, Rekognition, SNS, SQS, Elasticsearch, API Gateway; video analysis, search, and download for surveillance product.
- **FOTA / iDVR Billing Report API** — Node.js, CloudWatch, S3, Athena, Lambda, DynamoDB, API Gateway; firmware-over-the-air + fleet management billing.
- **Operations tooling** — automated email delivery (SES + S3 templates); CloudWatch alert configuration tool; cross-cloud monitoring → Google Chat webhook push (Lambda + SNS + Ansible).
- **ISO 27001 Service Monitor** — CloudWatch dashboard for compliance audit.

SKILLS

Category	Technologies
AWS	EKS · ECS · RDS · ElastiCache · VPC · IAM Identity Center · Cost Explorer · Organizations · ECR · SSM · Lambda · Route53 · ACM · CloudWatch · EventBridge · SNS · Secrets Manager · WAF · GuardDuty · Security Hub · Config · Athena · Client VPN · KMS · Rekognition
GCP	GKE · Cloud SQL · Cloud Functions · IoT Core · BigQuery · Cloud Armor · Security Command Center · Cloud Monitoring · PubSub · PSC · DMS
IaC	Terraform (3-layer state, S3 + DynamoDB lock) · AWS CDK (TypeScript / Python) · Helm · Kustomize
Kubernetes	EKS · GKE · k3s · ArgoCD · KEDA · HPA / VPA · Istio · External Secrets Operator · IRSA · Velero · cert-manager
CI/CD	GitLab + GitLab Runner · Jenkins (JNLP + Kaniko + S3 / EFS cache) · GitHub Actions (reusable workflows, OIDC)
Observability	Grafana · OpenTelemetry · Thanos · ELK / OpenSearch · PMM · CloudWatch · Cloud Monitoring · Sentry
Security & Compliance	SOC 2 · ISO 27001 · WAF · GuardDuty · Security Hub · Cloud Armor · cert-manager · Network Policy · Chaos Monkey · DefectDojo + Dependency-Track
AI / DevTools	LLM-based SRE automation (OpenClaw) · Claude Code multi-agent skill platform
Languages	Python · TypeScript · Node.js · Bash · HCL · YAML
Other	Fortinet VPN · IoT (MQTT, GPIO, Raspberry Pi) · AWS Client VPN